指数函数 → 矩阵指数

$y(x) = e^{ax}$

~~$y'(x) = ae^{ax}$~~

$$\begin{cases} y'(x) = a y(x) \\ y(0) = 1 \end{cases}$$

假设可展开成幂级数

$y(x) = c_0 + c_1 x + c_2 x^2 + \cdots$

代入 $x=0$ $y(0) = c_0 = 1$

$y'(x) = c_1 + 2c_2 x + 3c_3 x^2 + \cdots$

$a y(x) = a + ac_1 x + ac_2 x^2 + \cdots$

$3c_3 = ac_2$

$$\frac{c_3}{c_2} = \frac{a}{3}$$

$c_0 = 1$ $c_1 = a$ $c_2 = c_1 \frac{c_2}{c_1} = a \cdot \frac{a}{2}$

$c_3 = c_1 \cdot \frac{c_2}{c_1} \frac{c_3}{c_2} = a \cdot \frac{a}{2} \cdot \frac{a}{3}$

$$c_n = \frac{a^n}{n!}$$

$$e^{ax} = 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \cdots$$

$$\begin{cases} y' = ay \\ y(0) = 1 \end{cases} \Rightarrow y = e^{ax} = \sum \frac{(ax)^k}{k!}$$

求满足方程

$$\frac{dy}{dx} = ay$$

两边积分: $\frac{dy}{y} = a\,dx$

$$\ln|y| = ax + C_1$$

通解 $y = e^{ax+C_1} = Ce^{ax}$

代入 $y(0) = 1$ $\quad Ce^0 = 1 \quad C = 1$

特解: $y = e^{ax}$

$y = y(x)$

变成 $x = x(t)$ 时变量

标量

$\dot{x} = qx$

~~$x(t=0)$~~

~~$t=0\ x(0)=x_0$~~ 错

$\frac{dx}{dt} = qx$

$\frac{dx}{x} = q\,dt$

积分

$\ln|x| = at + C$

通解 $x = Ce^{at}$

限定 $x(t_0) = x_0$

令 $at_0 = 0$

代入 $x(0) = x_0$

~~$t=0$~~

$x_0 = C \cdot e^{a\cdot 0} = C$

$\therefore C = x_0$

$x = x_0 e^{at}$

$x_0 = Ce^{at_0}$

$C = x_0 \cdot e^{-at_0}$

代回

$x = C \cdot e^{at}$

$= x_0 \cdot e^{a(t-t_0)}$

又 $e^{at} = 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} + \cdots$

$= \sum_{k=0}^{\infty} \frac{a^k}{k!} t^k$

类似的

$x(t)$ 从标量 → $x(t)$ 向量

a → 方阵 A

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{A^k}{k!} t^k$$

幂级数.

$$\exp[A] = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

e^{At}

等例1

$$e^M = \sum_{k=0}^{\infty} \frac{(M)^k}{k!}$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

e^{-At} 令 $M = -At$

$$-At = -t\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix}$$

$$M^2 = \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix}\begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -t^2 & 0 \\ 0 & -t^2 \end{bmatrix} = -t^2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -t^2 I$$

$$M^3 = -t^2 M$$

$$M^4 = t^4 I$$

$$M^5 = t^4 M$$

$$e^{-At} = e^M = I + M + \frac{M^2}{2!} + \frac{M^3}{3!} + \frac{M^4}{4!} + \cdots$$

$$= I + M + \frac{-t^2 I}{2!} + \frac{-t^2 M}{3!} + \frac{t^4 I}{4!} + \frac{t^4 M}{5!} + \cdots$$

$$= I\left(1 + \frac{-t^2}{2!} + \frac{t^4}{4!} + \frac{-t^6}{6!} + \cdots\right)$$

$$+ M\left(1 + \frac{-t^2}{3!} + \frac{t^4}{5!} + \frac{-t^6}{7!} + \cdots\right)$$

$$= I \cdot \cos t + M \cdot \frac{1}{t} \cdot \sin t$$

$$= \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

标量 e^{at} 求导 ae^{at}

矩阵 e^{At} 求导 Ae^{At}

$$e^{At} = I + At + \frac{(At)^2}{2!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}$$

证明如下

$$\frac{d}{dt} e^{At} = 0 + A + A^2 t + \frac{A^3 t^2}{2!} + \cdots = \sum_{k=1}^{\infty} \frac{A^k t^{k-1}}{(k-1)!}$$

$$= 0 + A + \frac{A^2 t}{1!} + \frac{A^3 t^2}{2!} + \frac{A^4 t^3}{3!} + \cdots$$

0! 为 1 为了让 $n! = n \times (n-1)!$ 在成立

令 $m = k-1$

$$\frac{d}{dt} e^{At} = \sum_{m=0}^{\infty} \frac{A^{m+1} t^m}{m!} = A \cdot \sum_{m=0}^{\infty} \frac{A^m t^m}{m!}$$

$$= A \cdot e^{At}$$

暫时不去深究

泰勒展开 $\rightarrow$ 七个公式泰勒展开.

三角函数泰勒展开 四个相等.

二阶四阶无需.

~~e^{At} 怎么~~

~~求这个东西用最简单的~~

$$\Phi(t,t_0) = e^{A(t-t_0)}$$

这里写 $t_0=0$ 情况，用 t 换成 $t-t_0$ 即可

e^{At}

①

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots$$

②

$$\dot{x} = Ax$$

L：~~$sX(s) = AX(s)$~~

$$sX(s) - x_0 = AX(s)$$

~~$x_0 = (sI - A)X(s)$~~

$$(sI - A)X(s) = x_0$$

$$X(s) = (sI - A)^{-1}x_0$$

L^{-1}：

$$x(t) = L^{-1}[(sI - A)^{-1}]x_0$$

③ 特征对角化求法，

求特征值 $A=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$

$(\lambda I - A)=\begin{bmatrix}\lambda & 1\\ -1 & \lambda\end{bmatrix}$

$\det(\lambda I - A)=0$

$\lambda^2+1=0$

$\lambda_1=j \quad \lambda_2=-j$

求每个特征值对应特征向量

$(\lambda I - A)\vec{v}_i=\vec{0}$

$\begin{bmatrix}j & -1\\ 1 & j\end{bmatrix}\begin{bmatrix}v_{11}\\ v_{12}\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$

$v_{11}j-v_{12}=0$ 取 $\vec{v}_1=\begin{bmatrix}1\\ j\end{bmatrix}$

$\begin{bmatrix}-j & -1\\ 1 & -j\end{bmatrix}\begin{bmatrix}v_{21}\\ v_{22}\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$ $\vec{v}_2=\begin{bmatrix}1\\ -j\end{bmatrix}$

$P=[\vec{v}_1,\vec{v}_2]=\begin{bmatrix}1 & 1\\ j & -j\end{bmatrix}$

P^{-1} =

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}^{-1} = \frac{1}{-2j}\begin{bmatrix} -j & -1 \\ -j & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}$$

对角线矩阵 $\tilde{A} = P^{-1}AP$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}$$

$$= \begin{bmatrix} \frac{j}{2} & \frac{1}{2} \\ -\frac{j}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} j & -j \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix}$$

验算 $P\tilde{A}P^{-1} = A$?

$$\begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}\begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix}\begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}$$

$$= \begin{bmatrix} j & -j \\ -1 & -1 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$ ✓

先记

$$e^{PDP^{-1}} = P \cdot e^{D} \cdot P^{-1}$$

$$e^{PDP^{-1}} = I + (PDP^{-1}) + \frac{(PDP^{-1})^2}{2!} + \frac{(PDP^{-1})^3}{3!} + \cdots$$

$$PDP^{-1} \cdot PDP^{-1} = P \cdot D^2 \cdot P^{-1}$$

$$= P\left(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \cdots\right)P^{-1}$$

$$= P \cdot e^{D} \cdot P^{-1}$$

D 这是个矩阵可以提成 A, At, $-At$

$$P^{-1}AP = D = \hat{A}t$$

$$e^{PDP^{-1}} = e^{P\hat{A}tP^{-1}} = e^{At}$$

$$e^{At} = P \cdot e^{\hat{A}t} \cdot P^{-1}$$

$A\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $\tilde{A} = \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix}$

~~$e^{j\theta} = \sin\theta + j\cos\theta$~~

$\tilde{A}t = jt$

~~$jt = e^{jt}$~~

$e^{\tilde{A}t} = e^{\begin{bmatrix} jt & 0 \\ 0 & -jt \end{bmatrix}}$

D为对角阵 $D = \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix}$ $D^2 = \begin{bmatrix} d_1^2 & 0 \\ 0 & d_2^2 \end{bmatrix}$ ...

$e^D = I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \cdots$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} + \begin{bmatrix} d_1^2 & 0 \\ 0 & d_2^2 \end{bmatrix} + \cdots$

$= \begin{bmatrix} 1+d_1+d_1^2+\cdots & 0 \\ 0 & 1+d_2+d_2^2+\cdots \end{bmatrix} = \begin{bmatrix} e^{d_1} & 0 \\ 0 & e^{d_2} \end{bmatrix}$

$e^{\begin{bmatrix} jt & 0 \\ 0 & -jt \end{bmatrix}} = \begin{bmatrix} e^{jt} & 0 \\ 0 & e^{-jt} \end{bmatrix}$

$$P = \begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}$$

$$\tilde{\Lambda}_t = \begin{bmatrix} e^{jt} & 0 \\ 0 & e^{-jt} \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}$$

$$P\tilde{\Lambda}_t P^{-1} = \begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix} \begin{bmatrix} e^{jt} & 0 \\ 0 & e^{-jt} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}$$

$$= \begin{bmatrix} e^{jt} & e^{-jt} \\ je^{jt} & -je^{-jt} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{j}{2} \\ \frac{1}{2} & \frac{j}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}e^{jt} + \frac{1}{2}e^{-jt} & -\frac{j}{2}e^{jt} + \frac{j}{2}e^{-jt} \\ \frac{1}{2}je^{jt} - \frac{1}{2}je^{-jt} & \frac{1}{2}e^{jt} + \frac{1}{2}e^{-jt} \end{bmatrix}$$

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{jt} = \cos t + j\sin t$

$$= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

$\frac{1}{2}(\cos t + j\sin t + \cos t - j\sin t)$

$= \cos t$

$\frac{j}{2}(-\cos t - j\sin t + \cos t - j\sin t)$

$= \sin t$